

THE PROBLEM

Short electrical transients can make a pulse-output sensor report events that never happened. A low event count during an expected operating window can also signal a process fault. PulseTrust gives an integrator a compact, auditable digital front end.

WHAT THE CIRCUIT DOES

Synchronize a conditioned logic-level input with two flip-flops; filter transitions until 1-8 stable clock samples arrive; count accepted rising edges up to 255; then snapshot each host-defined measurement window with glitch, low-count, overflow and ready flags.

PIN CONTRACT

SIGNAL	FUNCTION
ui[0]	Raw conditioned digital pulse input
ui[3:1]	Stable samples minus one: 0-7 means 1-8 samples
ui[4], [5], [6]	Enable, synchronous clear, window boundary
ui[7]	Read live count or last completed count
uio_in[7:0]	Minimum expected pulses in the closing window
uo_out[7:0]	Selected saturating 8-bit count
uio_out[0:6]	Filtered level, accepted strobe, glitch, low-count, overflow, ready, enable

LIVE PROOF

A seeded synthetic input with 5 intended events and 8 short spikes produced 13 raw edges. The actual Verilog simulation accepted 5 and rejected 8. A self-checking Icarus test passes noise rejection, valid counting, window snapshots, low-count alerts and overflow; five Cocotb tests pass in GitHub Actions, including gate-level tests. The Tiny Tapeout GDS and 1x1 precheck jobs passed: 284 mapped standard cells, 14.76% utilization, zero final route DRC and LVS errors. The timed flow reports +10.95 ns slow-corner setup slack at a 20 ns constraint.

ENGINEERING LIMITS

A filter can reject narrow valid pulses; the fastest setting can accept noise. At 50 MHz, 1-8 samples cover only about 20-160 ns: a mechanical reed switch would need a slower supplied clock or a prescaler. The count saturates at 255. Synchronization reduces, but cannot eliminate, metastability risk. An input protection stage and host control are external. The physical metrics are tool-flow evidence, not measured silicon. No fabricated chip or certified measurement is claimed.

SIGNAL PATH

